# SOLUTION

## SECTION–I

**1. Ans. (C)**

**Sol.** E : none of the drawn balls are green

$E_1$ : 2R balls are drawn

$E_2$ : 2W balls are drawn

$E_3$ : 1R + 1W balls are drawn

$$P(E_1/E)$$

$$=\frac{P(E_1).P(E/E_1)}{P(E_1)P(E/E_1)+P(E_2).P(E/E_2)+P(E_3).P(E/E_3)}$$

$$=\frac{\frac{{}^3C_2}{{}^7C_2}\times 1}{\frac{{}^3C_2}{{}^7C_2}\times 1+\frac{{}^2C_2}{{}^7C_2}\times 1+\frac{{}^3C_1\times{}^2C_1}{{}^7C_2}}=\frac{3}{10}$$

**2. Ans. (C)**

**Sol.** $P(A)-P(A\cap B)=\frac{1}{3}$ &

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)=\frac{11}{15}$$

$$\Rightarrow \quad P(B)=\frac{6}{15}=\frac{2}{5}$$

$$P(A)-P(A)\,P(B)=\frac{1}{3}\Rightarrow P(A)=\frac{5}{9}$$

$$\Rightarrow \quad P(A\cap B)=P(A)\,P(B)=\frac{2}{5}\times\frac{5}{9}=\frac{2}{9}$$

**3. Ans. (B)**

**Sol.** $\because \quad \{P\}=P \Rightarrow P\in[0,1)$

$\Rightarrow P\neq 1$

$P(E)\neq 1 \Rightarrow$ E is not a sure event.

**4. Ans. (C)**

**Sol.** $\because\ b=\frac{7}{9}a$

$\therefore$ (a,b) ≡ (9,7); (18,14); (27,21); (36,28)

$$\therefore \text{ Required probability } =\frac{4}{{}^{39}C_2}=\frac{4}{741}$$

**5. Ans. (D)**

**Sol.** All throws are independent

$$\therefore\ P=\frac{1}{6}$$

**6. Ans. (C)**

**Sol.** $P(A\cap\bar{C})=P(A\cap\bar{C}\cap B)+P(A\cap\bar{C}\cap\bar{B})$

$$=\frac{1}{3}+\frac{1}{3}=\frac{2}{3}$$

$$P(A)=P(A\cap C)+P(A\cap\bar{C})$$

$$\Rightarrow\ P(A\cap C)=\frac{3}{4}-\frac{2}{3}=\frac{1}{12}$$

**7. Ans. (A)**

**Sol.** P(i) = ki

$$\because \quad \sum_{i=1}^{6}P(i)=1 \Rightarrow\ k(1+2+3+4+5+6)=1$$

$$\Rightarrow\ k=\frac{1}{21}$$

$\therefore$ P(even) × P(odd)

$$=\left(\frac{2}{21}+\frac{4}{21}+\frac{6}{21}\right)\times\left(\frac{1}{21}+\frac{3}{21}+\frac{5}{21}\right)=\frac{12}{49}$$

**8. Ans. (B)**

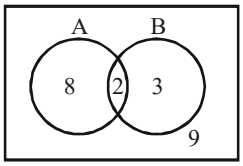
**Sol.** P(WBWB) + P(BWBW)

$$=\frac{5}{8}\times\frac{3}{8}\times\frac{5}{8}\times\frac{3}{8}+\frac{3}{8}\times\frac{5}{8}\times\frac{3}{8}\times\frac{5}{8}$$

$$=2\left(\frac{15}{64}\right)^2=\frac{15^2}{2^{11}}$$

**9. Ans. (B,C,D)**

**Sol.** $P(\bar{A})=\frac{6}{11};P(A)=\frac{5}{11}$




$$P(B)=\frac{5}{22},\ P(A\cap B)=\frac{1}{11}$$

$$P(A\cup B)=\frac{13}{22},\ P(\bar{A}\cap\bar{B})=\frac{9}{22}$$

$$P(\bar{A}\cup\bar{B})=\frac{10}{11},\ P(\bar{B}\cap A)=\frac{8}{22}=\frac{4}{11}$$

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**10. Ans.(A,B,C,D)**

**Sol.** (A) $P(\bar{A}\cup\bar{B}) = P(\overline{A\cap B}) = 1-P(A\cap B)$

$1 - P(A)\,P(B) = \frac{5}{6}$

(B) $= P(B) - P(B\cap C) = P(B) - P(B)P(C)$

$= \frac{5}{18}$

(C) $= P(\overline{A\cup B}) = 1-P(A\cup B)$

$= 1-P(A)-P(B)+P(A\cap B)$

$= 1-P(A)-P(B)+P(A)P(B) = \frac{1}{3}$

(D) $= P(B) - P(A\cap B) + P(C) - P(B\cap C) + P(A) - P(A\cap C)$

$= P(B) - P(A)\,P(B) + P(C) - P(B)\,P(C) + P(A) - P(C)\,P(A) = \frac{25}{36}$

**11. Ans.(A,B,C,D)**

**Sol.** (A) $P(A\cup B\cup C) = 1-P(\bar{A}\cap\bar{B}\cap\bar{C})$

$1-\frac{2}{3}.\frac{1}{2}.\frac{3}{4} = \frac{18}{24} = \frac{3}{4}$

(B) $\Sigma P(AB) - 3P(ABC) = \frac{1}{6}+\frac{1}{8}+\frac{1}{12}-\frac{3}{24}$

$= \frac{4+3+2-3}{24} = \frac{1}{4}$

(C) $\Sigma P(AB) - 2P(ABC) = \frac{1}{6}+\frac{1}{8}+\frac{1}{12}-\frac{2}{24} = \frac{7}{24}$

(D) $1-P(A\cap B\cap C) = 1-\frac{1}{3}\times\frac{1}{2}\times\frac{1}{4} = \frac{23}{24}$

**Paragraph for Question 13 to 14**

**13. Ans. (C)**

**Sol.** $P(B) = \frac{5}{6}p_1 + \frac{5}{6}(1-p_1)(1-p_2).\frac{5}{6}p_1 + \left(\frac{5}{6}\right)^3 (1-p_1)^2(1-p_2)^2 p_1 + \ldots\ldots$

$P(B) = \frac{\frac{5}{6}p_1}{1-\frac{5}{6}(1-p_1)(1-p_2)}, P(C) = \frac{\frac{5}{6}(1-p_1)p_2}{1-\frac{5}{6}(1-p_1)(1-p_2)}$

$\frac{P(B)}{P(C)} = \frac{p_1}{p_2(1-p_1)}$

**14. Ans.(C)**

**Sol.** If game is equally likely

$\Rightarrow$ Probability becomes

$\frac{1}{3}$ for each player $\Rightarrow \frac{P(B)}{P(C)} = 1$

$\Rightarrow p_1 = p_2(1-p_1)$ ...(i)

$\& \; P(A) = \frac{\frac{1}{6}}{1-\frac{5}{6}(1-p_1)(1-p_2)} = \frac{1}{3}$

$\Rightarrow (1-p_1)(1-p_2) = \frac{3}{5}$ ...(ii)

solve (i) & (ii) $\Rightarrow p_2 = \frac{1}{4}, p_1 = \frac{1}{5}$

$\Rightarrow p_1 : p_2 = 4:5$

**Paragraph for Question 14 & 15**

**14. Ans. (A)**

**Sol.** P(student will pass) $= \frac{1}{7}.\frac{1}{2}+\frac{1}{3}.\frac{1}{3}+\frac{11}{21}.\frac{1}{6} = \frac{17}{63}$

**15. Ans. (C)**

**Sol.** P(y/ student is fail) $= \frac{\frac{1}{3}.\frac{2}{3}}{\frac{1}{7}.\frac{1}{2}+\frac{1}{3}.\frac{2}{3}+\frac{11}{21}.\frac{5}{6}} = \frac{7}{23}$

## SECTION–II

**1. Ans. 2.00**

**Sol.** Prime numbers

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29}

P(atleast one) = 1 – P(no prime)

$= 1-\frac{{}^{20}C_2}{{}^{30}C_2} = 1-\frac{20.19}{30.29} = \frac{49}{87}$

$\therefore \quad \frac{b-a}{19} = \frac{87-49}{19} = 2$

**2. Ans. 106.00**

**Sol.** Two balls transferred from I to II can be 1W and 1B or both W or both B.

Then the required probability

$= \left(\frac{{}^4C_1\,{}^3C_1}{{}^7C_2}\right).\frac{6}{9} + \left(\frac{{}^4C_2}{{}^7C_2}\right)\frac{7}{9} + \left(\frac{{}^3C_2}{{}^7C_2}\right).\frac{5}{9}$

$= \frac{43}{63} = \frac{p}{q} \quad \therefore\; p+q = 106$


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**3. Ans. 4.00**

**Sol.** Possible points that Humpty Not scores are

$$= \begin{cases} 9 & \text{i.e.} \quad 333 \\ 8 & \text{i.e.} \quad 233 \\ 7 & \text{i.e} \quad 223 \end{cases}$$

Required probability = 1–P(he score 7,8 or 9)

$= 1-\{{}^3C_2(0.3)^2.(0.5)+{}^3C_2.(0.5)^2.(0.3)+{}^3C_3.(0.5)^3\}$

$$= 1-\left\{\frac{3.9.5+3.25.3+1.125}{1000}\right\} = \frac{103}{200}$$

**4. Ans.** 0.49

**Sol.** $E_r$ : hunters shoots the animal at r distance

$P(E) = P(\text{escapes}) = 1 - P(\bar{E}_{2a}\bar{E}_{3a}.....\bar{E}_{na})$

$$= 1-\left(1-\frac{a^2}{4a^2}\right)\left(1-\frac{a^2}{9a^2}\right)....\left(1-\frac{a^2}{n^2a^2}\right)$$

$$= 1-\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)....\left(1-\frac{1}{n^2}\right)$$

$$= 1-\left(\frac{1.3}{2.2}\right)\left(\frac{2}{3}.\frac{4}{3}\right)\left(\frac{3}{4}.\frac{5}{4}\right)....\left(\frac{n-1}{n}\frac{n+1}{n}\right) = 1 -$$

$$\frac{n+1}{2n} = \frac{n-1}{2n} = \frac{49}{100} = 0.49$$

## SECTION–IV

**1. Ans. (A)→(P); (B)→(Q,T); (C)→(P); (D)→(S,T)**

**Sol.** (A) Let number of black balls = x

$$\Rightarrow \quad \frac{{}^2C_1.{}^2C_1+{}^2C_1.{}^xC_1+{}^2C_1.{}^xC_1}{{}^{x+4}C_2} = \frac{5}{7}$$

$$\Rightarrow \quad 7(4+4x) = \frac{5(x+4)(x+3)}{2}$$

$\Rightarrow$ $56 + 56x = 5x^2 + 35x + 60$

$\Rightarrow$ $5x^2 - 21x + 4 = 0$

$\Rightarrow$ $(5x - 1)(x - 4) = 0 \quad \Rightarrow \quad x = 4$

(B) $\because$ A,B are independent

$\Rightarrow$ $A, \bar{B}$ are also independent

$$\Rightarrow \quad P(A/\bar{B}) = P(A) = \frac{1}{3}$$

(C) 1 – P(1 on none of the throw) > 0.5

$$\Rightarrow \quad 1-\left(\frac{5}{6}\right)^n > 0.5 \Rightarrow \quad \left(\frac{6}{5}\right)^n > 2$$

$\Rightarrow$ $n_{min} = 4$

(D) $P(3H \ \& \ 5T) = {}^8C_3 \times \left(\frac{1}{2}\right)^3 \times \left(\frac{1}{2}\right)^5$

$$= 56 \times \frac{1}{8 \times 32}$$

$$\Rightarrow \quad \frac{7}{32}$$

$\Rightarrow$ odds in favour = 7 to 25

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